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$\gcd(a, b, c) = d$ st $d|a$, $d|b$, & $d|c$, d is greatest

a

let k be any positive integer

$k|s \Rightarrow k|b$ & $k|c$ by def of $\gcd$

$$\gcd(a, s) = 1 \Rightarrow 1|a \text{ \& } 1|s \Rightarrow 1|a \text{ \& } 1|b \text{ \& } 1|c$$

def of $\gcd$ also states that 1 is the greatest num to divide a & s

thus 1 is the greatest number that divides $a, b, \& c$, $\gcd(a, b, c)$

b
$$\gcd(a, \gcd(b, c)) = \gcd(a, b, c)$$

by bezout's theorem $\gcd(b, c) = bm + cn$

for some $m, n \in \mathbb{Z}$

again by bezout's theorem

$$\begin{aligned} \gcd(a, bm + cn) &= au + v(bm + cn) \quad v = bt \quad w = ct \\ &= au + bv + cw \quad \text{for some } u, v, w \in \mathbb{Z} \end{aligned}$$

2 a

$$n = m^2 \quad n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} \quad m = p_1^{b_1} p_2^{b_2} \dots p_j^{b_j}$$

as $n = m \cdot m$, no prime can be in the factorization of n without being in the factorization of m , thus $k = j$

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} = (p_1^{b_1} p_2^{b_2} \dots p_k^{b_k})^2$$

by exponent rules,

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} = p_1^{2b_1} p_2^{2b_2} \dots p_k^{2b_k}$$

by the fundamental theorem of arithmetic

$$a_i = 2b_i \quad 1 \leq i \leq k$$

as $a_i \geq 2$, all exponents in the prime factorization of n must be even or 0

b

if n is a perfect square, $\sqrt{n} = m$, and is thus rational, as shown in a if we instead assume m is rational, i.e. $m = \frac{p}{q}$ $\gcd(p, q) = 1$ then to show $n = m^2 \in \mathbb{Q}$

$$n = \frac{p^2}{q^2} \text{ decomposing into prime factors } n = \frac{(p_1^{a_1} p_2^{a_2} \dots p_k^{a_k})^2}{(q_1^{b_1} q_2^{b_2} \dots q_j^{b_j})^2}$$

No p_i can be equal to any q_i , as then $\gcd(p, q) \neq 1$

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$$n = (p_1^{2a_1} p_2^{2a_2} \dots p_k^{2a_k}) (q_1^{2b_1} q_2^{2b_2} \dots q_j^{2b_j})$$

$2a_i$ are all even
 $2 \cdot (-b_i)$ are all even
 thus by part a, n is a perfect square

3 a $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R} \quad f(x, y) = (x+y, x-y)$
 injective: yes surjective: yes

b $g: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z} \quad g(x, y) = (x+y, x-y)$
 injective: yes surjective: no $(1, 0)$ is not possible

4 a criteria for equivalence relation
 reflexive
 symmetric
 transitive

$\frac{a}{b} \sim \frac{a}{b} \Rightarrow ab = ab \quad \checkmark \quad \sim \text{is reflexive}$

$\frac{a}{b} \sim \frac{c}{d} \Rightarrow ab = cd \quad \& \quad \frac{c}{d} \sim \frac{a}{b} \Rightarrow cd = ab$

normal equality is symmetric, so is $\sim$

$\frac{a}{b} \sim \frac{c}{d} \quad \& \quad \frac{c}{d} \sim \frac{e}{f}$

$\Rightarrow ab = cd \quad \& \quad \Rightarrow cd = ef$

$\Rightarrow ab = ef \Rightarrow \frac{a}{b} \sim \frac{e}{f} \quad \sim \text{is transitive}$

$\sim$ is an equivalence relation

b if $\frac{a}{b}$ is not already a reduced fraction (ie $\gcd(a, b) \neq 1$)

$\frac{a}{b}$ can be converted into one as follows

$\gcd(a, b) = c \quad a = m \cdot c, \quad b = n \cdot c \quad m, n \in \mathbb{Z}$

$\frac{a}{b} = \frac{m \cdot c}{n \cdot c} = \frac{m}{n} \quad \gcd(m, n) = 1$

Sign can be handled as follows if $b < 0$, multiply by $\frac{-1}{-1}$, so that b is positive